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EmbryoGuard Pro

IoT-Based Smart Embryo Incubator System

CSE272: Embedded Systems Project Presentation

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Team Members

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Project Background

Embryo incubation is a critical process in poultry farming that requires precise environmental control to ensure optimal hatch rates.

Traditional incubators often lack real-time monitoring, remote access capabilities, and intelligent automation, making it difficult to maintain strict conditions.

The **EmbryoGuard Pro** system addresses these limitations by leveraging modern IoT technologies, embedded systems, and web-based dashboards to create a comprehensive, intelligent incubator management platform.



Core Objectives



Dual-Node IoT

Implement a robust dual-node system using an ESP32 microcontroller, utilizing FreeRTOS for concurrent monitoring and active control tasks.



Precise Control

Achieve exact temperature regulation using advanced hysteresis-based control logic, maintaining a highly accurate $\pm 0.5^{\circ}\text{C}$ tolerance band.



Real-Time Web

Provide a responsive, full-stack dashboard with bidirectional WebSocket communication for instant metric updates and remote actuation.

System Architecture

Firmware Layer

Built on a dual-core architecture running on a single ESP32.

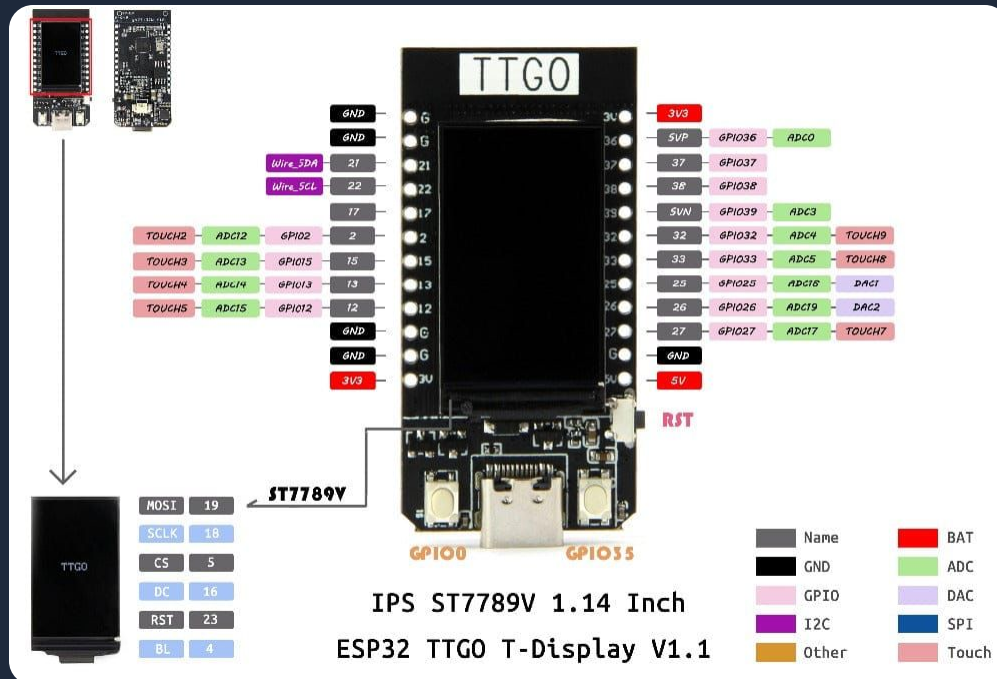
Core 1 manages active climate control, controls actuator toggling (fans/lamps), and handles command polling. **Core 0** handles independent environmental monitoring and HTTP data posting.

Application Layer

Powered by a robust Python Flask web server paired with SocketIO for bidirectional data streaming.

Features a persistent SQLite database for extensive sensor logging and a responsive HTML/JS dashboard interface for remote user control.

Hardware Design

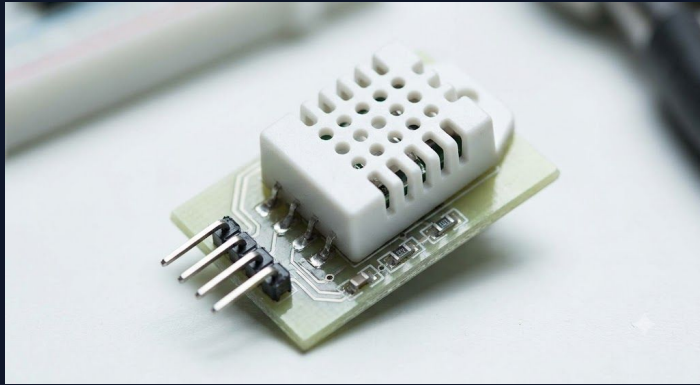


ESP32 Microcontroller

Dual-core processor designed for robust IoT applications.

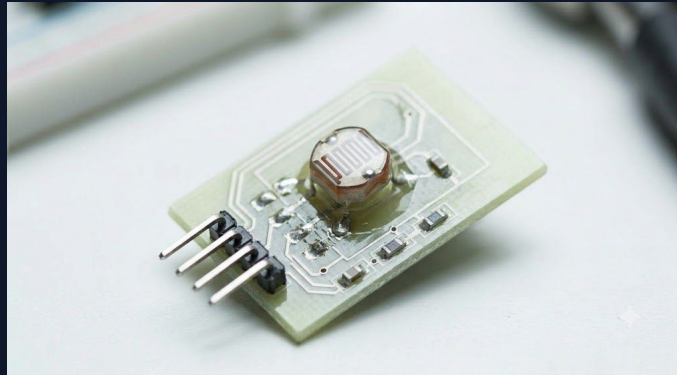
- High-speed Wi-Fi and Bluetooth connectivity.
- FreeRTOS integration for efficient task threading.
- Low power consumption with high performance.

Hardware Design



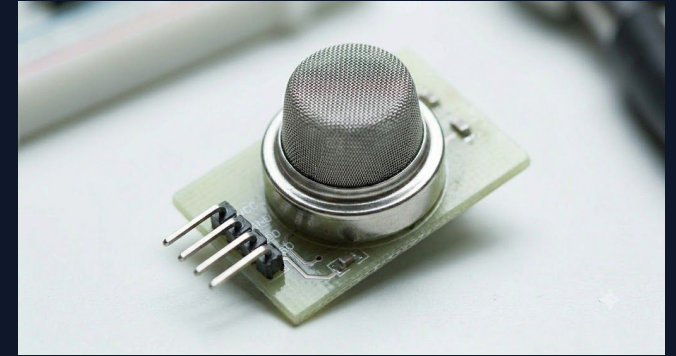
DHT22

Dual-function sensor for accurate climate monitoring (temperature and humidity).



LDR Module

Analog light-dependent resistor for detecting shifts in ambient lighting.



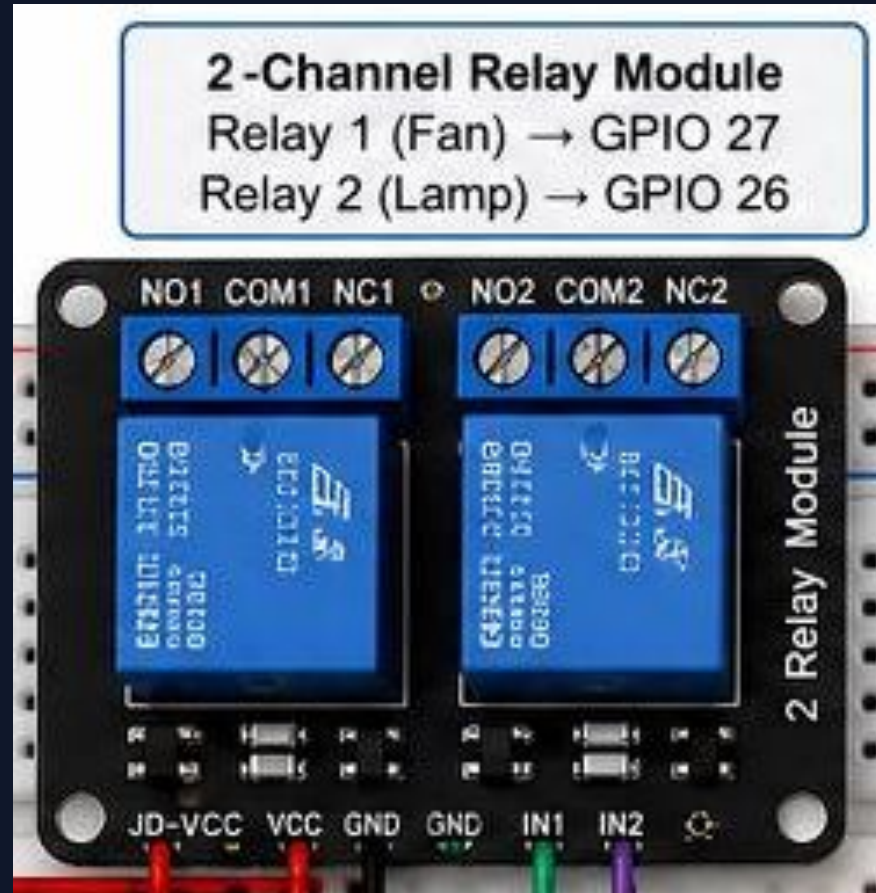
MQ-135

Broad-spectrum gas sensor designed for comprehensive air quality analysis.

Hardware Design



Hardware Design



Hardware Design



Hardware Design



Hardware Design

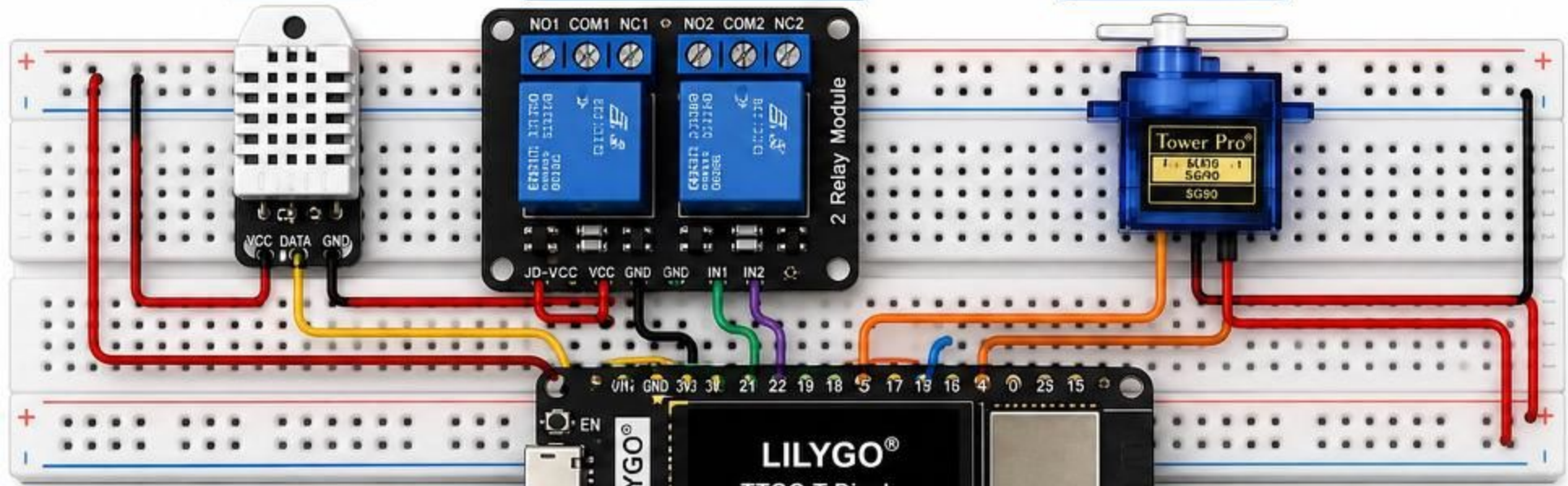


NODE 1

DHT22
(GPIO 21)

2-Channel Relay Module
Relay 1 (Fan) → GPIO 27
Relay 2 (Lamp) → GPIO 26

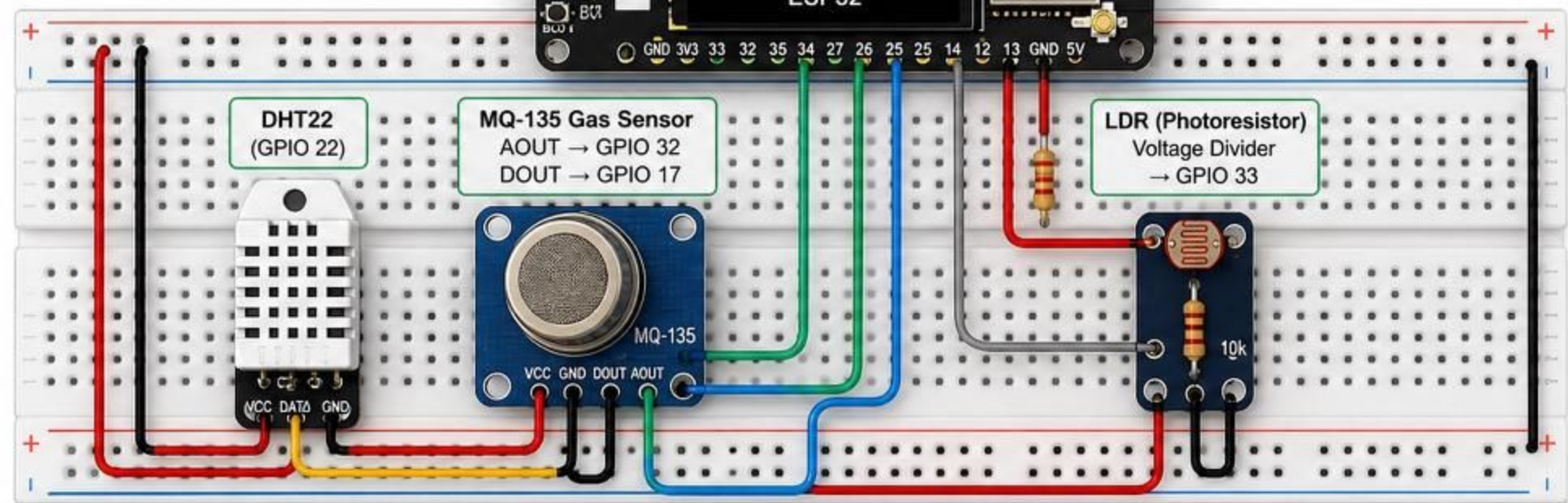
SG90 Servo Motor
(GPIO 13)



DHT22
(GPIO 22)

MQ-135 Gas Sensor
AOUT → GPIO 32
DOUT → GPIO 17

LDR (Photoresistor)
Voltage Divider
→ GPIO 33



PIN CONNECTION SUMMARY

DHT22 (Node 1)	Yellow	→ GPIO 21
Relay 1 (Fan)	Green	→ GPIO 27
Relay 2 (Lamp)	Purple	→ GPIO 26
SG90 Servo	Orange	→ GPIO 13
DHT22 (Node 2)	Yellow	→ GPIO 22
MQ-135 DOUT	Green	→ GPIO 17
MQ-135 AOUT	Blue	→ GPIO 32
LDR (Voltage Divider)	Grey	→ GPIO 33

POWER

3.3V / 5V	Red	→ Red Wires
GND	Black	→ Black Wires

NOTES

- All components share common GND.
- Use 3.3V for sensors (DHT22, MQ-135, LDR).
- Use 5V for Relay Module (VCC & JD-VCC) and Servo Motor.

Hysteresis Algorithm

±0.5°C

Precision Target Range

Unyielding Climate Stability

The auto-regulation system utilizes a tight hysteresis band around the target temperature to prevent actuator oscillation.

For a standard **37.5°C** profile:

- **Heating (Lamp ON):** Activates at 37.0°C
- **Cooling (Fan ON):** Activates at 38.0°C

This guarantees optimal embryo development without exposing eggs to harmful thermal extremes.

Multi-Species Profiles

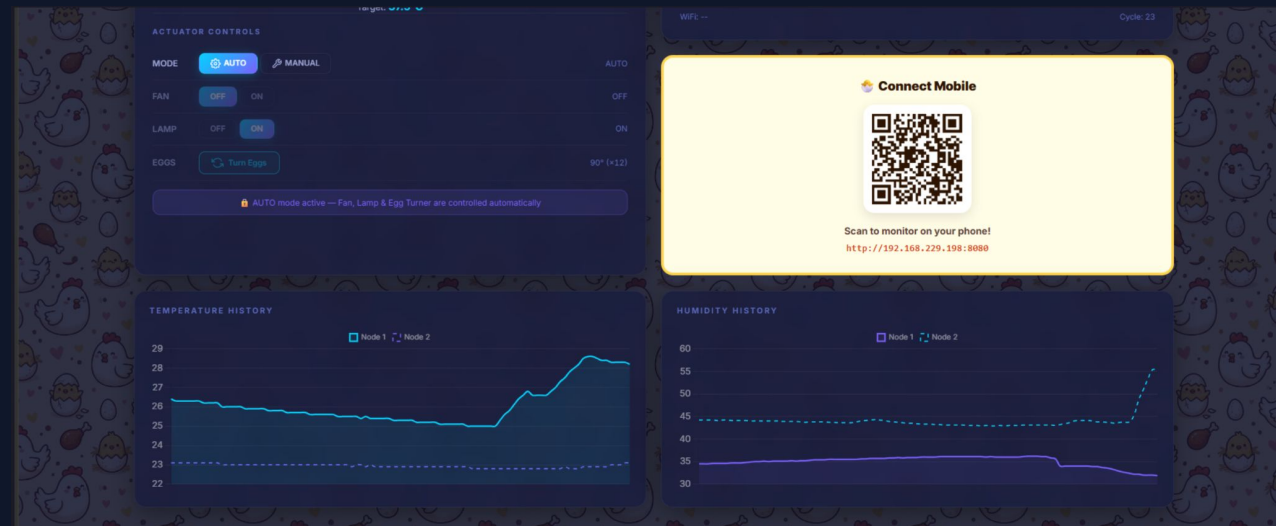
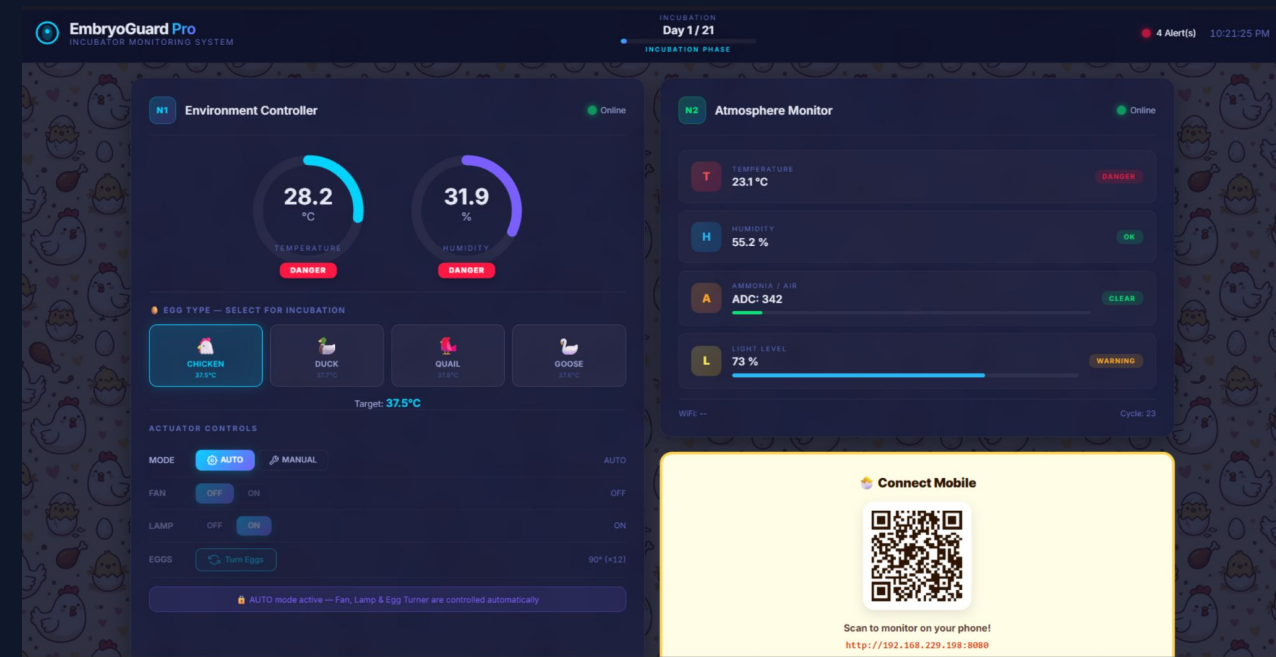
Species	Target Temp (°C)	Lamp ON Below (°C)	Fan ON Above (°C)	Incubation Days
Chicken 🐔	37.5	36.5	38.5	21
Duck 🦆	37.7	37.0	39.0	28
Quail 🐣	37.8	37.5	39.5	18
Goose 🦢	37.6	36.8	38.8	25

Dashboard Interface

Real-Time Control Hub

The custom web dashboard, built with Flask and SocketIO, acts as the central command center, providing instant bidirectional communication with the ESP32 hardware.

It features real-time circular gauges, dynamic threshold updates, and interactive actuator controls. Operators can actively monitor live Chart.js graphs, track incubation progress, and evaluate intelligent alerts from any device.



Safety & Alert System



DANGER OVERRIDE: Extreme temperatures ($\geq 39.5^{\circ}\text{C}$ or $\leq 35.0^{\circ}\text{C}$) trigger immediate emergency actuator states, aggressively bypassing any active manual commands.



FAIL-SAFE SHUTDOWN: Total sensor failure (NaN readings) automatically causes all heating and cooling actuators to shut down to prevent unintended thermal runaway.



WATCHDOG TIMER: A 10-second hardware timer (esp_task_wdt) continuously monitors FreeRTOS tasks and forcefully resets the system in the event of firmware hangs.



TIMEOUT REVERT: To prevent operator error, manual control mode safely auto-reverts to automatic climate regulation after 15 minutes of inactivity.

Questions?

Thank you for your attention.

EmbryoGuard Pro Team | Supervised by Dr. Safa Elaskary